01 TAKE HOME TEST — DATA ANALYST

What Makes Deliveries Late?

Zomato Delivery Operations Analytics

PRESENTER

Vino Jayrasta Samudra

Traffic | Multi-Order | Festival

02 Brief Profile

Self-overview, education, and work experience

SELF-OVERVIEW

- **Psychology → Data Analytics:** combining an understanding of human behavior with data patterns.
- **Data → Insight → Action:** turning data and visualizations into executable plans.
- **Multi-Perspective Analysis:** more accurate and balanced decisions.
- **International Career Vision:** a data analyst at international scale, building data-driven strategies for long-term growth.

Bachelor of Psychology
Universitas 17 Agustus 1945 Surabaya

GPA
3.68

WORKING EXPERIENCE

Marketing Staff · PT Surya Trias Gemilang

August 2022 – August 2026

20+	Prospective customers reached per month via WhatsApp, email, Google Ads, and the website
50+	Transactions managed end-to-end, from quotation to closing — total net profit of **Rp 200,000,000**
50+	Transactions and customer accounts analyzed to identify trends and support management decisions
20+	SEO-friendly product content and promotional materials — website traffic and inquiries up **30%**
90%	On-time delivery rate across 50+ unit deliveries, coordinated with the operations team

03 Project Overview

Four portfolio projects from the bootcamp

PROJECT 01

RFM Segmentation

Superstore · 793 customers · $2.3M sales

Goal: segment customers for targeted marketing strategies.

Analysis: RFM model producing 11 segments.

Result: At Risk was the largest sales contributor (20.4%) despite long recency.

Excel | Power BI

PROJECT 02

A/B Testing

Spotify · trending song recommendation feature

Goal: measure the feature's impact on engagement, payment rate, and listening time.

Analysis: t-test and z-test, 50,000 users per group.

Result: all 3 metrics rose significantly ($p<0.001$), H0 rejected, full rollout recommended.

Google Colab (Python)

PROJECT 03

Marketing Channel Analysis

Marketing channel evaluation

Goal: evaluate channel effectiveness to optimize the marketing budget.

Analysis: dashboard with KPI cards and a heatmap matrix per channel.

Result: all channels were nearly identical on ratio metrics.

Power BI

PROJECT 04

Customer Segmentation (Final Project)

DataCo Supply Chain · 20,652 customers

Goal: identify customers who cause losses for the company.

Analysis: RFM scoring → 3 segments (At Risk, Regular, High Value).

Result: 19.7% of customers are At Risk, with a **$1.08M** loss.

Power BI

04 Project Background

Context, goal, and why this project matters

Analyzing **40,100 Zomato delivery records** to find the factors that most affect **late food deliveries**.

Main Goal

Identify the **operational and external factors** that make orders late.

Expected Outcome

A list of **actionable drivers** of late deliveries for management.

Why It Matters

23% of orders are late; delivery time affects customer satisfaction and loyalty.

Who Benefits

Zomato operations management, couriers, and customers.

05 Business Problem

23% of orders arrive late

Total Orders	Avg Delivery Time	Late Delivery Rate	Avg Distance
40,100	26.3 min	23%	9.7 km

PROBLEM STATEMENT

"Which factors most affect late deliveries, and which operational conditions can be optimized?"

OBJECTIVES

01 Identify the **drivers** of late deliveries

02 Compare **late rates** across conditions

03 Build **data-driven recommendations** for management

06 Data Understanding

Source, structure, and data quality

Source

Zomato Delivery Operations Analytics Dataset (Kaggle).

Structure

45,584 rows × **20** columns, **11 Feb – 6 Apr 2022**.

Courier, restaurant, location, weather, traffic, vehicle, and delivery time data.

Data Quality

- 228–1,908 missing values per column
- 91 dummy rows (age 15 & rating 1.0 / age 50 & rating 6.0)
- Negative and zero coordinates
- Mixed time formats
- 0 duplicates

Final Dataset

Cleaning kept most of the data, and the **average delivery time stays 26.3 min** before and after cleaning.

Rows kept	Data retained	Avg delivery time
40,100	88%	26.3 min

07 Preprocessing & Feature Engineering

Cleaning steps, new features, and how "late" is defined

STEP 01

Missing Values

Missing categories → "Unknown". Missing multiple_deliveries → 0 (checked: averages unchanged).

STEP 02

Coordinates

Negative coordinates fixed (sign error). Zero coordinates removed.

STEP 03

New Features

distance_km (Haversine), **order_hour**, **prep_time**.

STEP 04

Outliers

231 outliers (>51 min) **kept**, because they are valid and have clear causes.

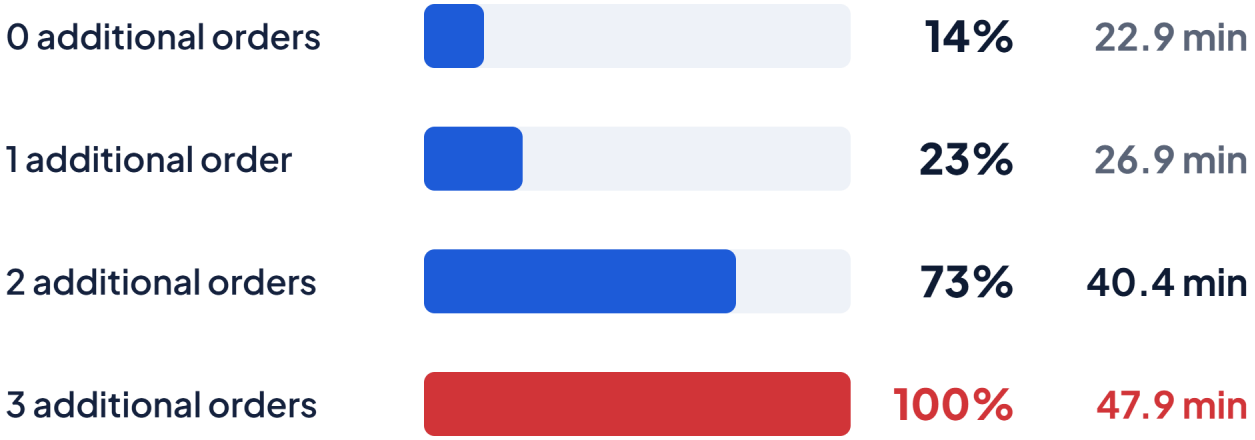
Late Definition

An order is late if it falls in the **slowest 25%** of orders within its distance group.

0–5 km	5–10 km	>10 km
> 27 min	**> 29 min**	**> 38 min**

08 More Orders per Trip, More Late Deliveries

Insight 1 · Multi-order: late rate and average delivery time by additional orders per trip



Bar = % of orders late within each group · Avg time = average delivery time

HOW MUCH SLOWER?

+25 min

Average delivery time more than **doubles**: 22.9 min with no extra orders vs **47.9 min** with 3.

The biggest jump is from 1 to 2 extra orders: late rate **23%** → **73%**, delivery time **+13.5 min**.

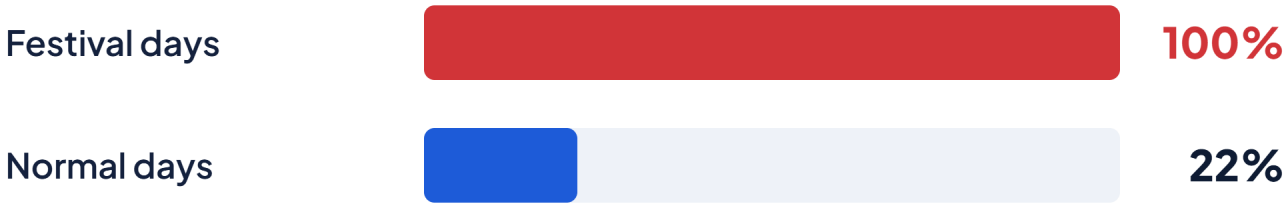
09 All Deliveries During Festivals Were Late

Insight 2 · Festival: late rate and average delivery time, festival days vs normal days

AVERAGE DELIVERY TIME

Festival days	Normal days
45.5 min	**26.0 min**

LATE DELIVERY RATE



HOW MUCH SLOWER?

+19.5 min

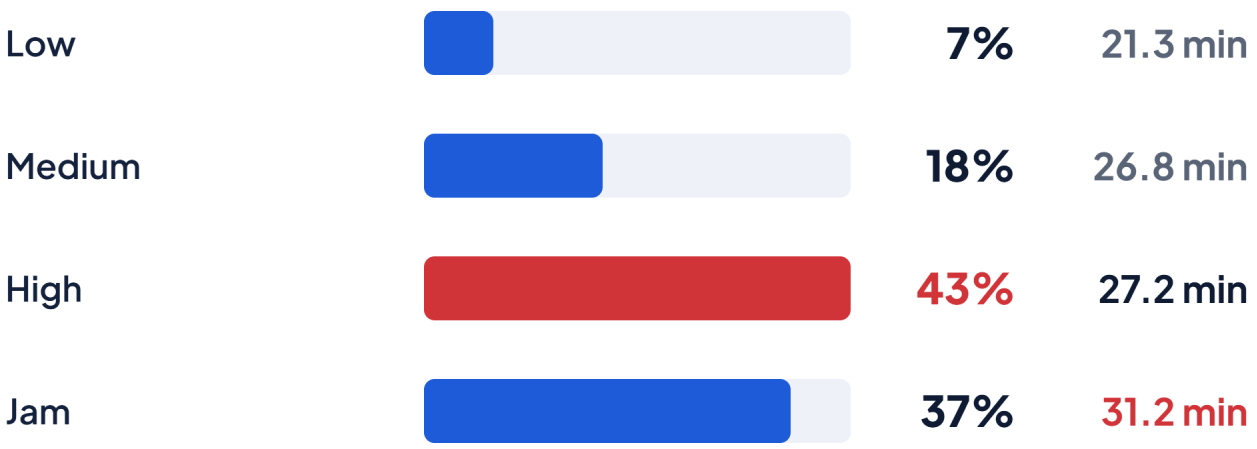
Even the **fastest** festival delivery took 38 min, above every late threshold.

⚠ **Note:** the festival sample is **783 orders** (~2% of the data).

10 Heavy Traffic Increases Late Deliveries up to 6×

Insight 3 · Traffic & order hour: late rate and average delivery time by traffic density

LATE DELIVERY RATE BY TRAFFIC



Bar = % of orders late · Avg time = average delivery time (Jam is +9.9 min vs Low)

SAME DISTANCE: HIGH ≈ JAM

49% vs 49%

Late rate for High vs Jam traffic at **0–5 km**. Jam has the longest average time because most Jam orders are long-distance.

ORDER HOUR · AVG TIME

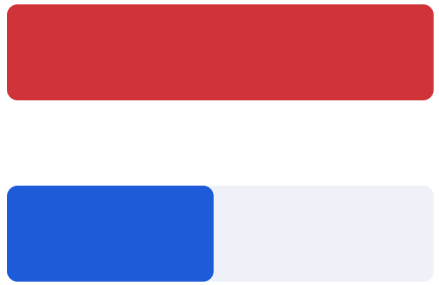
Slowest · 19–21	~31 min
Fastest · 8–10	~19.5 min

Note: in this dataset, traffic is determined by order hour.

11 Poor Vehicle Condition Doubles the Late Rate

Insight 4 · Vehicle condition: late rate and average delivery time by vehicle condition

LATE RATE BY VEHICLE CONDITION



Bar = % of orders late · Avg time = average delivery time (+5.7 min for condition 0)

THE SLOWEST DELIVERIES

All 231

of the slowest deliveries (**>51 min**) used **condition-0** vehicles.

⚠ VEHICLE TYPE TRAP

Motorcycles look slower (**27% vs 17%**) only because every condition-0 vehicle is a motorcycle.

12 Distance and Cloudy/Fog Weather Add Delays

Insight 5 · Distance & weather: average delivery time and late rate

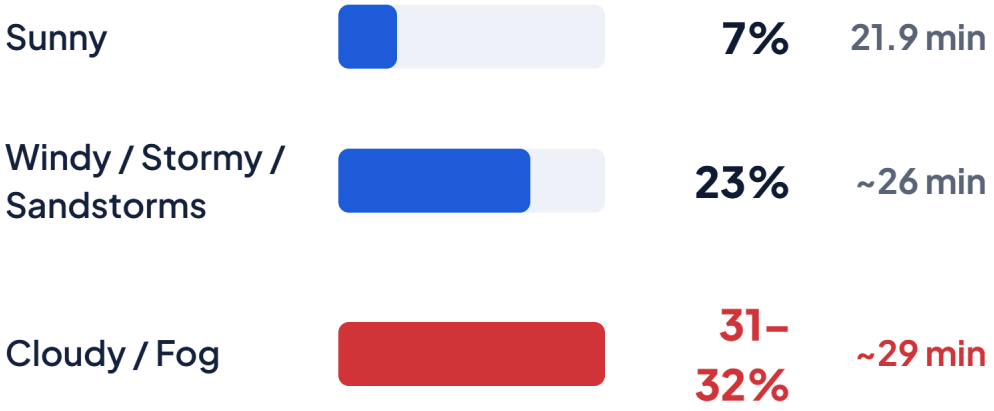
DISTANCE · AVG DELIVERY TIME

1–9 km	**21–25 min**
>10 km	**~30 min**

After 10 km, delivery time stays flat.

Correlation with delivery time **0.32**

WEATHER · LATE RATE & AVG TIME



Bar = % of orders late · right = average delivery time (+7.1 min vs Sunny)

13 Courier Rating and Age Also Matter

Other factors: two moderate correlations, two with no effect

Courier Rating

Correlation

−0.36

Rating ≤4.0	**36 min**
Rating ≥4.6	**~24.5 min**

⚠ **Caution:** rating may be a result, not a cause.

Courier Age

Correlation

0.30

Age 20–30	**~23–24 min**
Age 31–39	**~29.5 min**

⚠ Not used as a decision basis.

No Effect

Restaurant prep time · correlation

−0.01

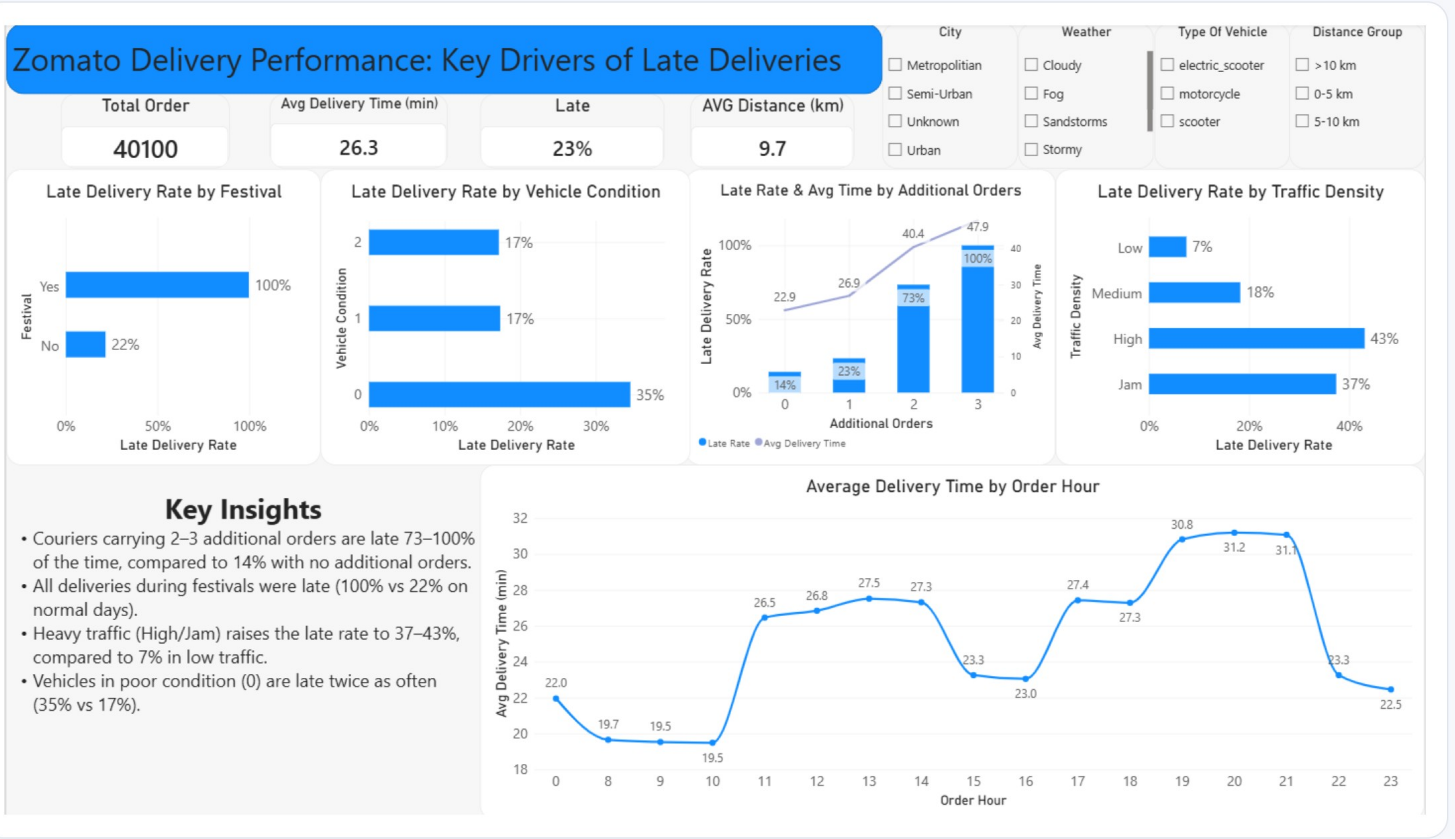
Order type · difference

1%

Kitchen speed and order type do not explain late deliveries.

14 Power BI Dashboard

Interactive dashboard for exploring late deliveries



FEATURES

- **4** KPI cards
- **5** charts, incl. late rate + avg time combo
- **Filters:** City, Weather, Vehicle, Distance Group
- **Key Insights** panel

15 Recommendations

Five actions for management, each with a KPI to track

SHORT TERM

Limit multi-order

Max 1 additional order per trip during peak hours (19–21) and festivals.

KPI · % of orders with ≥2 additional orders

Dynamic ETA in the app

Add a buffer for >10 km orders, peak hours, festivals, and Cloudy/Fog weather.

KPI · Gap between estimated and actual delivery time

Smarter courier assignment

Assign >10 km orders to couriers with good-condition vehicles and no additional orders.

KPI · Late rate for >10 km orders

MEDIUM TERM

Vehicle maintenance program

Maintain or replace condition-0 vehicles.

KPI · % of fleet in condition 0 · late rate of condition-0 vehicles

Festival capacity plan

Add couriers on scheduled festival days.

KPI · Late rate during festivals

Start with multi-order: 73% of trips with 2 additional orders and 100% with 3 are late.

16 Data Limitations

What to keep in mind when reading these results

- **Data is likely synthetic:** prep time only 5/10/15 min, courier age evenly spread 20–39, step-shaped delivery times.
- **Traffic** is determined by order hour.
- **Distance** is a straight line, not the road route.
- **No official SLA:** the late threshold was built from the data.
- **Semi-Urban** has only 138 orders.

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Thank You

Questions & discussion

PRESENTER
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